PV102 Day 2: Site and Solar Resource Assessment

Students convert vague site assumptions into observable site conditions: solar access, shade, orientation, tilt, space, weather exposure, access, roof or ground constraints, and maintenance realities.

PV102 | CTS Day 18 | Instructor Run Sheet
Measure the site before you size the system. | Tuesday, August 4, 2026

Core rule

Measure the site before you size the system.

Learning targets

- Identify site factors that affect PV production and installation practicality.
- Use orientation, tilt, shade, and access observations to support a design decision.
- Separate measured observations from assumptions and preferences.
- Create a site-note handoff that Day 3 can use for load and energy work.

Readiness and materials

Before students arrive	Materials
Open slide deck and test display. Print student packets and one instructor key. Prepare three team stations. Review safety boundary before any equipment is handled.	Compass or phone compass app Tape measures Clipboards Site sketch sheets Markers Scenario cards Optional shade objects: cardboard/tree props/ladder silhouette Projector and slide deck

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	Opening and continuity	Connect Day 1 to today's rule. Name what students already know and what new judgment is being added.
10:15-10:40	Morning quiz and review	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:25	Concept lesson	Use slides 1-14. Pause after each technical rule and ask what decision it changes.
11:25-12:15	Guided practice	Model one example on the board. Students complete a quick pair-check before lunch.
12:15-12:45	Lunch/reset	Reset lab stations. Keep energized work out of scope unless explicitly authorized by instructor and equipment procedure.

12:45-1:00	Lab brief	Assign teams and roles. Review acceptance criteria and stop-work triggers.
1:00-2:15	Site Walk and Solar Access Handoff	Teams inspect three candidate locations and decide which one deserves further design work.
2:15-2:40	Handoffs and peer verification	Each team gives a concise technical handoff. Peers check evidence, boundary, and unknowns.
2:40-3:00	Exit ticket and reset	Collect packets. Preview next step: PV102 Day 3 uses the site handoff to build a load profile and first-pass energy budget.

Instructor talk track

Opening script: "PV102 is not about throwing parts at a problem until reality files a complaint. Today we use site and solar resource assessment to make the next design decision safer and more defensible. The rule is simple: Measure the site before you size the system."

Teaching notes

Move	Look-for
Ask	What decision does this evidence change?
Listen for	Students separating facts, assumptions, and unknowns.
Correct gently	Component-first answers. Redirect to job, evidence, boundary, then component.
Require	A handoff that another person could act on without reading minds. Humanity keeps trying telepathy. It remains a poor project-management system.

Lab instructor procedure

- Assign one candidate location per team.
- Sketch the site from above and mark north, shade sources, access path, approximate array area, and likely wire route.
- Record observations in three columns: measured, observed, unknown.
- Name the best reason to keep or reject the site for first-pass design.
- Prepare a two-minute handoff for Day 3.

Answer snapshot and contingency plan

If equipment or printing is unavailable, run the lab with paper cards, chart paper, and oral handoffs. The learning target is decision quality, not hardware theater.

Safety and scope boundaries

No live electrical work is authorized by this lesson unless the instructor independently sets up a verified training system, follows manufacturer instructions, and establishes the required isolation, PPE, and supervision. Treat PV source circuits and batteries as energy sources, not props.

Day handoff

PV102 Day 3 uses the site handoff to build a load profile and first-pass energy budget.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Department of Energy, Solar Photovoltaic Technology Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-technology-basics
- U.S. Department of Energy, Solar Photovoltaic System Design Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-system-design-basics
- U.S. Department of Energy, Solar Integration: Solar Energy and Storage Basics: https://www.energy.gov/eere/solar/solar-integration-solar-energy-and-storage-basics
- U.S. Department of Energy, Solar Integration: Inverters and Grid Services Basics: https://www.energy.gov/eere/solar/solar-integration-inverters-and-grid-services-basics
- NREL PVWatts Calculator: https://pvwatts.nrel.gov/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- OSHA, Green Job Hazards - Solar Energy: Personal Protective Equipment: https://www.osha.gov/green-jobs/solar/personal-protective-equipment
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.